

COURSE OVERVIEW

Module 1: Introduction to Python

- What is Python and history of Python?
- Unique features of Python
- Python-2 and Python-3 differences
- Install Python and Environment Setup
- First Python Program
- Python Identifiers, Keywords and Indentation
- Comments and document interlude in Python
- Command line arguments □ Getting User Input
- Python Data Types
- What are variables?
- Python Core objects and Functions
- Number and Maths

Module 2: Control Statements

- if-else
- if-elif-else
- while loop
- for loop
- break
- continue
- assert
- pass
- return



Module 3: List, Ranges & Tuples in Python

- Introduction
- Lists in Python
- More about Lists
- Understanding Iterators
- Generators, Comprehensions and Lambda Expressions
 - Introduction
 - Generators and Yield
 - Next and Ranges
- Understanding and using Ranges
- More About Ranges
- Ordered Sets with tuples

Module 4: Python Dictionaries and Sets

- Introduction to the section
- Python Dictionaries
- More on Dictionaries
- Sets
- Python Sets Examples

Module 5 : Input and Output in Python

- Reading and writing text files
- writing Text Files
- Appending to Files and Challenge
- Writing Binary Files Manually
- Using Pickle to Write Binary Files

Module 6 : Python built in function

- Python user defined functions
- Python packages functions
- Defining and calling Function
- The anonymous Functions
- Loops and statement in Python
- Python Modules & Packages



Module 7 : Python Object Oriented

- Overview of OOP
- The self variable
- Constructor
- Types Of Variables
- Namespaces
- Creating Classes and Objects
- Inheritance
- Types of Methods
 - Instance Methods
 - Static Methods
 - Class Methods
- Accessing attributes
- Built-In Class Attributes
- Destroying Objects
- Abstract classes and Interfaces
- Abstract Methods and Abstract class
- Interface in Python
- Abstract classes and Interfaces

Module 8: Exceptions

- Errors in Python
- Compile-Time Errors
- Runtime Errors
- Logical Errors
- What is Exception?
- Handling an exception
- try....except...else
- try-finally clause
- Argument of an Exception



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- Python Standard Exceptions
- Raising an exceptions
- User-Defined Exceptions

Module 9: Python Regular Expressions

- What are regular expressions?
- The match Function
- The search Function
- Matching vs searching
- Search and Replace
- Extended Regular Expressions
- Wildcard

Module 10: Python Multithreaded Programming

- What is multithreading?
- Difference between a Process and Thread
- Concurrent Programming and GIL
- Uses of Thread
- Starting a New Thread
- The Threading Module
- Thread Synchronization
 - Locks
 - Semaphore
- Deadlock of Threads
- Avoiding Deadlocks
- Daemon Threads



Module 11: Using Databases in Python

- Python MySQL Database Access
- Install the MySQLdb and other Packages
- Create Database Connection
- CREATE, INSERT, READ Operation
- DML and DDL Oepration with Databases

Module 12: Data Science Using Python

- **Numpy:**
 - Introduction to numpy
 - Creating arrays
 - Indexing Arrays
 - Array Transposition
 - Universal Array Function
 - Array Processing
 - Array Input and Output
- **Pandas:**
 - What are pandas?
 - Where it is used?
 - Series in pandas
 - Index objects
 - Reindex
 - Drop Entry
 - Selecting Entries
 - Data Alignment
 - Rank and Sort
 - Summary Statics
 - Index Hierarchy
 - Matplotlib: Data Visualization
 - Python for Data Visualization
 - Welcome to the Data Visualization Section
 - Introduction to Matplotlib



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TRAINING HIGHLIGHTS

- **Recording Access** shared to students on Learnomate App
- **Professional Resume building** by Industrial working mentors
- **Placement assistance/Job requirement notification support/HR contacts**
- **Training Certificate:** Receive a recognized certificate upon course completion
- **LinkedIn, Naukri.com Profile:** Enhance your online presence with professionally curated profiles.
- **Flexible Learning Options:** Choose between offline and online training to suit your schedule.
- **Interview Preparation, Mock Interviews:** Nail your interviews with our tailored preparation and mock interview sessions
- **Real-time Scenarios Explained:** Learn through practical examples to master real-world applications.
- **? Doubt Sessions:** Clarify your doubts through dedicated doubt-clearing sessions.

